

Training-Free Token Reduction on MotionBench

Benchmark Results for Eleven Methods across Two Backbones

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Abstract

This report presents benchmark results for eleven training-free video-LLM token-reduction methods on MotionBench (8,052 questions, 4,018 scoreable), evaluated on two backbones at a standardized 32 frames and 15% nominal token retention. Headline numbers: the Qwen3-VL-8B baseline reaches 62.52% against LLaVA-OV-7B’s 52.66%, a 9.86-point gap that holds across all six question categories. On LLaVA-OV seven of nine methods are accuracy-neutral, falling inside the ± 1.54 -point resolution floor, while two collapse to 36.78% and 38.20%. On Qwen3-VL, at identical settings, all ten portable methods lose accuracy, ranging from -0.35 to -6.55 , and eight of those losses are statistically significant — so the ranking does not transfer between backbones. A controlled experiment on the two collapsed methods shows their loss does not depend on which tokens are kept: selecting by attention, at random, or evenly across all frames scores within 1.1 points, while the same code path retaining every token recovers the baseline. On this backbone a 15% budget costs roughly 15 points regardless of selection rule. Results are reported overall and by category; methodology, verification, and limitations are in the appendices.

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1 Setup

Benchmark. MotionBench: 8,052 four-option multiple-choice questions over video clips. 4,034 carry an NA ground truth and are excluded, leaving **4,018 scoreable items** — the denominator for every accuracy figure below. Six question categories, of unequal size (Table 2). Scoring letter-matches A–D against ground truth; generation is greedy.

Backbones. **LLaVA-OV-7B** (SigLIP-so400m vision tower, Qwen2-7B LLM, mid-2024) and **Qwen3-VL-8B** (Qwen3VLForConditionalGeneration, a native architecture rather than a LLaVA derivative, late-2025).¹

Configuration. All methods run at 32 frames. Methods exposing a single retention parameter are set to 15%; those whose budget is fixed internally, or that select frames rather than tokens, keep published defaults. Comparison is therefore at a common *nominal* setting, not at equal compute — measured budgets range from 7.5% to 50% (Appendix A).

Reading the numbers. At $n = 4,018$ the binomial 95% confidence half-width is ± 1.54 points; smaller differences are not resolvable and are not ranked. * marks a difference significant by McNemar’s test on paired predictions ($\chi^2 \geq 3.84$, $p < 0.05$); paired counts are in Appendix B. Every figure here was recomputed from per-sample prediction records (Appendix A).

2 Results

2.1 Summary

Table 1: All methods on both backbones, sorted by Qwen3-VL result. * = significant vs. that backbone’s baseline. The dash marks a cell where the method is not implementable on that backbone.

Method	LLaVA-OV-7B		Qwen3-VL-8B	
	Overall	Δ	Overall	Δ
<i>Baseline</i>	<i>52.66</i>	—	<i>62.52</i>	—
PruneVID	38.20	−14.46*	62.17	−0.35
DyCoke	53.36	+0.70	61.85	−0.67
HoliTom	53.14	+0.47	60.33	−2.19*
MDP3	53.06	+0.40	59.66	−2.86*
FastV	36.78	−15.88*	59.01	−3.51*
VisionZip (ctx.) [‡]	—	—	58.81	−3.71*
STTM	51.72	−0.95	57.07	−5.45*
FlashVID	53.31	+0.65	56.65	−5.87*
VideoITG	52.86	+0.20	56.35	−6.17*
AIM	52.86	+0.20	55.97	−6.55*

[‡] Contextual half only — Qwen3-VL’s vision tower has no CLS token, so VisionZip’s dominant half is not implementable. Not the published method. VisionZip’s complete form runs on LLaVA-1.5-7B (Table 8). DyTo appears in Table 8: it is bound to a Vicuna backbone and runs on neither of these two.

¹Project records refer to a `qwen.1.5` template on LLaVA-OV. That is a prompt template, not a different LLM: `qwen.1.5` and `qwen.2` render byte-identical prompts here and the model is Qwen2-7B either way.

Table 1 contains the report’s main observation. Reading across rows rather than down columns, the ordering does not survive the change of backbone: PruneVID is the worst result on LLaVA-OV and the best on Qwen3-VL; FlashVID is indistinguishable from the backbone on LLaVA-OV and loses 5.87 points on Qwen3-VL. Every Qwen3-VL delta is negative; six of nine LLaVA-OV deltas are positive but none resolvably so. A ranking measured on one backbone does not transfer to the other.

2.2 Backbones

Table 2: Baseline accuracy overall and by category (% correct within category). AO = Action Order, CM = Camera Motion, LM = Location-related Motion, MR = Motion Recognition, MO = Motion-related Objects, RC = Repetition Count. Chance is 25%.

Backbone	Overall	AO	CM	LM	MR	MO	RC
LLaVA-OV-7B	52.66	40.5	45.2	55.5	57.0	71.2	23.8
Qwen3-VL-8B	62.52	46.1	63.1	65.0	67.3	79.0	33.8
Δ	+9.86	+5.6	+17.9	+9.5	+10.3	+7.8	+10.0
n	4,018	519	385	546	1,478	690	400

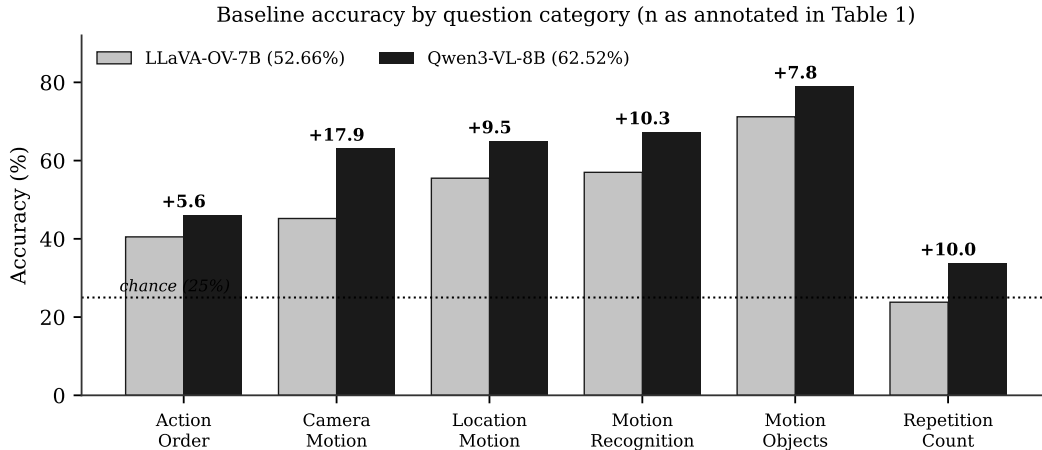


Figure 1: Baseline accuracy by category. Qwen3-VL-8B leads in all six, most sharply on Camera Motion (+17.9).

The 9.86-point backbone gap is larger than any method effect measured anywhere in this report, and it holds in every category.

Two properties of Table 2 govern how later tables should be read. Both backbones are strongest on Motion-related Objects (71.2%, 79.0%), the category most answerable from object appearance without integrating motion. And **Repetition Count is at or near chance for both** (23.8% and 33.8% against 25%): LLaVA-OV is not distinguishable from guessing there, so movement in that column is not evidence about a method and is not interpreted below.

2.3 LLaVA-OV-7B

Table 3: LLaVA-OV-7B, 32 frames, 15% nominal retention. All nine methods that run on this backbone. Categories as in Table 2. VisionZip runs on a different backbone (Table 8).

Method	Overall	Δ	AO	CM	LM	MR	MO	RC
DyCoke	53.36	+0.70	38.9	48.8	55.7	58.2	70.9	25.3
FlashVID	53.31	+0.65	39.9	45.7	53.8	58.0	71.7	28.2
HoliTom	53.14	+0.47	40.8	49.6	52.6	57.0	71.4	27.3
MDP3	53.06	+0.40	40.5	49.6	53.5	56.8	71.6	26.5
VideoITG	52.86	+0.20	40.1	47.0	53.7	57.6	70.1	26.8
AIM	52.86	+0.20	41.4	48.1	54.2	57.0	71.9	22.3
<i>Baseline</i>	<i>52.66</i>	—	<i>40.5</i>	<i>45.2</i>	<i>55.5</i>	<i>57.0</i>	<i>71.2</i>	<i>23.8</i>
STTM	51.72	−0.95	39.9	48.1	53.8	53.5	70.6	28.8
PruneVID-OV*	38.20	−14.46	32.0	34.8	35.5	38.4	52.9	27.3
FastV*	36.78	−15.88	32.8	31.9	34.1	35.7	53.8	25.3

Seven of the nine methods land within 0.95 points of the backbone, inside the ± 1.54 -point floor, and none is significant (largest $\chi^2 = 2.34$ against a 3.84 threshold). They are statistically indistinguishable from the backbone and from each other; the ordering among them is not meaningful. Their category profiles track the backbone closely, with the largest deviations falling in the two smallest categories. At a 15% budget on this backbone, seven of nine token-reduction methods are accuracy-neutral.

FastV and PruneVID-OV are the exceptions, losing more than 14 points each, significantly, and degrading in every category carrying above-chance backbone signal. §2.7 establishes what separates them from the other seven, and §2.8 isolates the cause by experiment.

2.4 Qwen3-VL-8B

Table 4: Qwen3-VL-8B, 32 frames, 15% nominal retention. All ten portable methods. VisionZip is the contextual half only (see Table 1 notes).

Method	Overall	Δ	AO	CM	LM	MR	MO	RC
<i>Baseline</i>	<i>62.52</i>	—	<i>46.1</i>	<i>63.1</i>	<i>65.0</i>	<i>67.3</i>	<i>79.0</i>	<i>33.8</i>
PruneVID	62.17	−0.35	46.1	62.9	64.1	67.1	79.1	32.5
DyCoke	61.85	−0.67	45.1	62.6	63.9	66.4	78.4	34.5
HoliTom*	60.33	−2.19	44.7	61.3	61.0	66.4	76.2	29.0
MDP3*	59.66	−2.86	44.9	64.7	62.8	64.0	77.4	23.0
FastV*	59.01	−3.51	46.2	61.0	61.9	63.5	72.2	30.5
VisionZip (ctx.)*	58.81	−3.71	43.2	57.4	61.9	64.2	74.3	29.5
STTM*	57.07	−5.45	44.3	60.0	61.0	60.6	73.8	23.8
FlashVID*	56.65	−5.87	43.4	57.9	57.1	61.2	73.0	27.0
VideoITG*	56.35	−6.17	45.3	53.5	59.3	60.2	75.4	22.2
AIM*	55.97	−6.55	45.9	56.6	59.5	58.3	72.2	27.0

Every method loses accuracy on Qwen3-VL and eight of ten lose significantly. The two exceptions, PruneVID (−0.35) and DyCoke (−0.67), are the two that retain roughly half their tokens against 7.5–15% for the rest (Appendix A).

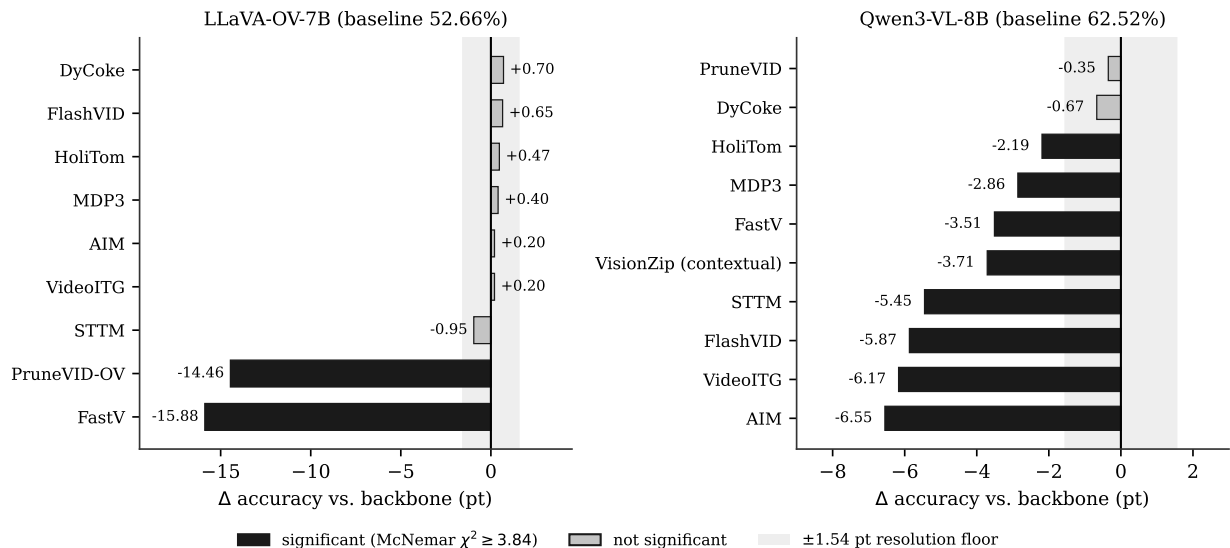


Figure 2: Change vs. backbone on each model. Shaded band is the ± 1.54 -point resolution floor. On LLaVA-OV every non-collapsed method falls inside it; on Qwen3-VL eight of ten fall outside it and all ten are negative.

Loss does not track budget monotonically. HoliTom operates at the smallest measured budget in the set (7.5%) and loses 2.19 points, while AIM and FlashVID at 15% lose 6.55 and 5.87. The methods that hold up are those that merge non-retained tokens into survivors rather than discarding them outright.

2.5 Per-category change

Losses are not spread evenly across question types. Expressing retained above-chance signal as $(\text{method} - 25) / (\text{backbone} - 25)$, FastV on LLaVA-OV keeps **62%** of the backbone’s above-chance signal on Motion-related Objects but only **30%** on Location-related Motion, **33%** on Motion Recognition and **34%** on Camera Motion. PruneVID-OV shows the same ordering (60% on Motion-related Objects, 34% on Location-related Motion). Motion-related Objects is the category most answerable from appearance in isolated frames; categories requiring integration across frames retain roughly half as much.

The pattern is milder but present on Qwen3-VL (Figure 3, right), where the largest losses fall on Motion Recognition (-9.0 , AIM) and Camera Motion (-9.6 , VideoITG), while Action Order moves within ± 3 points for every method.

Repetition Count is the control column. LLaVA-OV scores 23.8% there against 25% chance, so there is no signal to preserve; the apparent gains under FastV (+1.5) and PruneVID-OV (+3.5) accompany 15-point overall collapses and reflect answer redistribution (§2.7), not recovered capability.

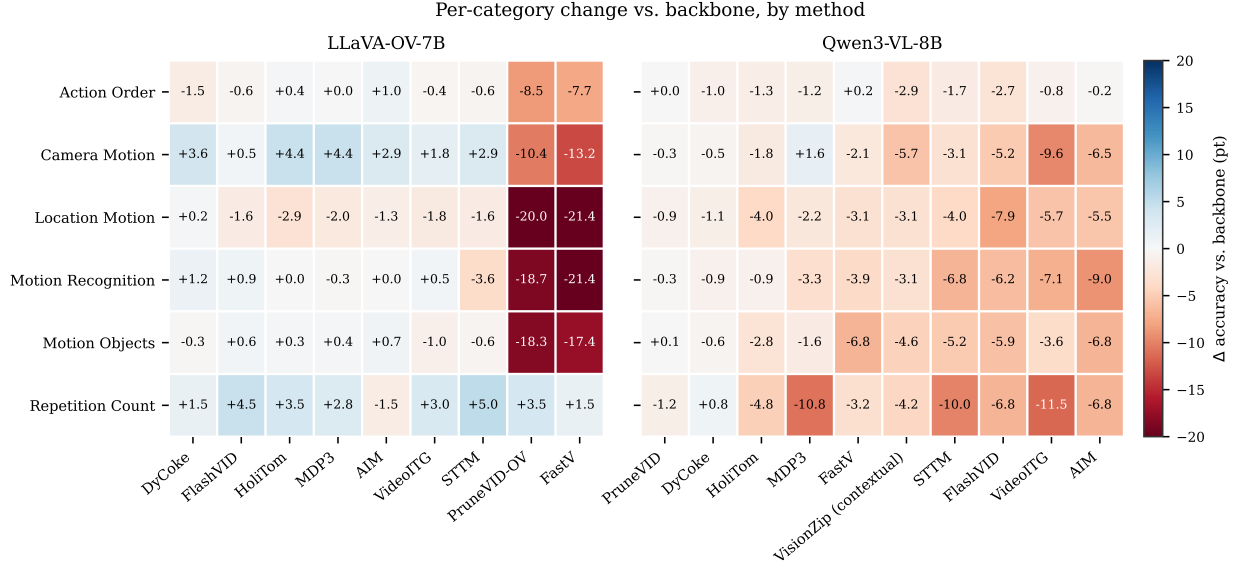


Figure 3: Per-category change vs. backbone, columns ordered by overall delta. Repetition Count movement is not interpretable — both backbones are at or near chance there (Table 2).

2.6 Retention sweep

Table 5: Accuracy (and Δ vs. backbone) across nominal retention, 32 frames and all else fixed. Span is movement from 10% to 25%. Four of eleven methods expose a comparable retention parameter.

Method	$r = 0.10$	$r = 0.15$	$r = 0.25$	Span
<i>LLaVA-OV-7B (baseline 52.66%)</i>				
FastV	36.06 (−16.60)*	36.78 (−15.88)*	36.73 (−15.93)*	+0.67
FlashVID	52.51 (−0.15)	53.31 (+0.65)	53.36 (+0.70)	+0.85
HoliTom	52.76 (+0.10)	53.14 (+0.47)	53.41 (+0.75)	+0.65
PruneVID	38.10 (−14.56)*	38.20 (−14.46)*	38.38 (−14.29)*	+0.28
<i>Qwen3-VL-8B (baseline 62.52%)</i>				
FastV	56.92 (−5.60)*	59.01 (−3.51)*	60.60 (−1.92)*	+3.68
FlashVID	54.73 (−7.79)*	56.65 (−5.87)*	59.36 (−3.16)*	+4.63
HoliTom	60.05 (−2.46)*	60.33 (−2.19)*	60.43 (−2.09)*	+0.38
PruneVID	60.23 (−2.29)*	62.17 (−0.35)	61.40 (−1.12)*	+1.17

Tripling the budget from 10% to 25% moves no LLaVA-OV method more than 0.85 points. On Qwen3-VL the same change moves FlashVID 4.63 points and FastV 3.68, with FastV’s significance decaying monotonically ($\chi^2 = 89.8 \rightarrow 44.6 \rightarrow 18.2$) as it converges toward the baseline — roughly a five-fold difference in retention sensitivity between the two backbones.

On LLaVA-OV the four methods sit in two flat regimes: FlashVID and HoliTom at the baseline at every setting, FastV and PruneVID roughly 15 points below it at every setting. Within this window, what costs accuracy is that those two discard tokens at all, not how hard. The window is bounded, however: §2.8 shows the same code path at 100% retention returns to the baseline, so the flatness across 10–25% locates a threshold below that range rather than showing budget to be irrelevant outright.

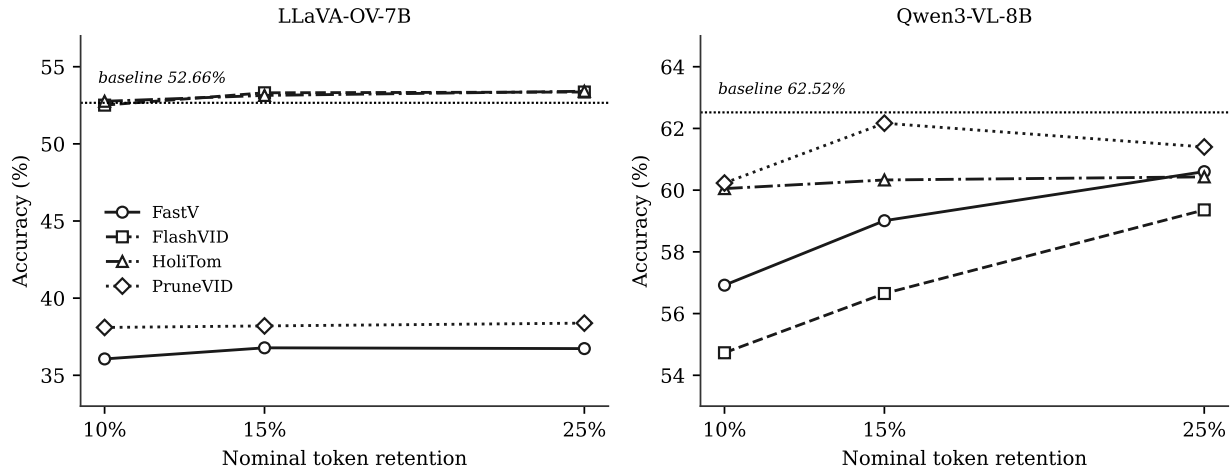


Figure 4: Accuracy against nominal token retention. On LLaVA-OV the four methods occupy two flat regimes the budget does not move; on Qwen3-VL the same methods respond strongly to the same parameter.

One cell is flagged rather than smoothed: PruneVID on Qwen3-VL is non-monotonic ($60.23 \rightarrow 62.17 \rightarrow 61.40$), its midpoint both its best result and its only non-significant one. At ± 1.54 -point resolution this may be noise; it is not reported as a trend.

2.7 Answer distribution on LLaVA-OV

Table 6: Distribution of predicted answer letters on LLaVA-OV-7B, % of 4,018 scoreable items.

Run	A	B	C	D
<i>Ground truth</i>	25.4	25.6	25.5	23.6
Baseline	27.1	26.6	26.9	19.3
DyCoke	27.4	26.1	27.0	19.3
FlashVID	27.7	25.6	25.7	20.8
HoliTom	26.1	24.1	27.2	22.5
PruneVID-OV	12.2	26.7	15.0	46.0
FastV	14.9	23.0	15.2	46.7

The two collapsed methods place 46–47% of their answers on option D against a 23.6% ground-truth rate, while the backbone and the three merge-based methods stay near uniform. This is the behaviour of a model answering from the prompt without usable visual evidence, and it explains the apparent Repetition Count gains in §2.5: a near-uniform category rewards a degenerate constant answer at roughly chance.

The signature is retention-invariant. Across $r \in \{0.10, 0.15, 0.25\}$ FastV’s D-rate is 47.0%, 46.8%, 47.1% — flat to a tenth of a point while accuracy moves 0.67 points. Restoring tokens does not restore visual grounding on this backbone, though it does for the same algorithm on Qwen3-VL, where the answer distribution tracks the backbone within 1.5 points at every budget.

Why two rows, not one. FastV and PruneVID-OV are different algorithms at different layers with different selection rules, yet fail with signatures matching to about one point on accuracy,

D-rate, broken/fixed ratio and flatness across retention. Two unrelated algorithms do not normally fail identically, and both LLaVA-OV ports drive reduction through the same code path — so the collapse could equally have been a property of that shared integration rather than of either method. §2.8 settles this by experiment.

2.8 What causes the collapse

Three explanations survived §2.7: the shared code path is defective; FastV’s attention ranking is uninformative at video scale; or the budget concentrates on too few frames. A four-arm experiment separates them. Each arm holds the backbone, prompt, frame count, budget and decoding fixed and varies only the token-selection rule applied at the pruning layer, on a 986-question subset whose category mix matches the full benchmark within 1.5 points (Table 7).

Table 7: Selection-rule experiment on LLaVA-OV-7B, 15% budget except where noted, $n = 986$. *Broke:fixed* is discordant pairs against the backbone on the same subset.

Arm	Selection rule	Overall	D-rate	Broke:fixed
<i>Baseline</i>	<i>no reduction</i>	<i>54.36</i>	<i>20.7</i>	—
keepall	same code path, discards nothing	55.07	20.8	0.36
attention	FastV as published	39.25	49.4	2.27
random	uniformly at random	38.74	47.2	2.43
uniform	evenly spaced across all 32 frames	38.13	49.9	2.43

The code path is sound. **keepall** traverses the identical reduction machinery while discarding nothing and returns 55.07% against a 54.36% subset baseline, with a near-baseline D-rate and a broke:fixed ratio below 1. The shared integration is not the cause, and both LLaVA-OV rows stand as measurements of the methods.

The attention ranking is inoperative. **attention** and **random** differ by 0.51 points, $\chi^2 = 0.16$. FastV’s layer-2 ranking performs no better than discarding 85% of visual tokens at random.

Frame coverage does not rescue it. **uniform** guarantees every frame contributes equally and is statistically indistinguishable from both other arms ($\chi^2 = 1.54$ and 0.30). Per-frame token histograms confirm the premise was wrong to begin with: every arm, **attention** included, touches all 32 frames on essentially every sample (mean 32.0 of 32), with the busiest frame taking 10.5% of **attention**’s budget against 3.1–4.2% for the others — far from concentration on a handful of frames.

What remains is a threshold effect. At a 15% budget on this backbone, discarding tokens costs about 15 points *regardless of which tokens survive*; at 100% the backbone is intact. Selection quality, and the temporal structure of the selection, stop mattering below that line. This is also the most economical account of why FastV and PruneVID-OV — unrelated in design — converge on nearly the same number: below the threshold there is nothing left for a selection rule to differentiate. Where exactly the threshold falls between 25% and 100% retention was not measured; the sweep in §2.6 covers only 10–25%, entirely below it (Appendix D).

The fork used here reproduces the production FastV run exactly on the matched subset (0 of 986 predictions differ), so the experiment measures the same object as Table 3.

2.9 Methods on non-standard backbones

Table 8: Runs on backbones other than the two standardized ones. Not comparable to Tables 3–4; no Δ is given because no matched baseline exists for either row.

Method	Backbone	Overall	AO	CM	LM	MR	MO	RC
VisionZip	LLaVA-1.5-7B	40.09	33.7	33.0	37.2	41.1	57.4	25.8
DyTo (recon. TW-FINCH)	Vicuna-7B	42.06	32.4	33.0	42.1	43.0	61.7	26.0

VisionZip patches `CLIPVisionTower`, which LLaVA-OV does not have, so its complete form runs only on LLaVA-1.5-7B — and at 8 frames rather than 32. The 40.09% differs from Table 3 in both backbone and frame count and is never placed in a LLaVA-OV comparison.

DyTo carries two caveats. It is not reproducible from its published artifacts (the released code calls an argument absent from its own pinned dependency, and a clustering helper has no return statement); the missing return was recovered by transcription from a character-identical sibling function, but the temporal weighting had to be reimplemented, so the result is reported only as “DyTo (reconstructed TW-FINCH)”. And it is the one number in this report with execution evidence but **no divergence check**, because no baseline was ever run on its Vicuna backbone — it is an absolute score, not a measured method effect.

A token-matched control is documented in the project records (cited, not recomputed here): against an uncompressed 6-frame baseline at 3,456 visual tokens scoring 43.35%, DyTo at $\sim$ 3,680 tokens scores 42.06%, a significant -1.29 . A frame-matched control is impossible — Vicuna’s 4,096-token context cannot hold 32 uncompressed frames — which is the constraint DyTo exists to address.

A Methodology and verification

The verification gate

A run that completes, reports a plausible accuracy, and did nothing is indistinguishable from a successful one by accuracy alone. Every number in this report therefore required two independent signals.

1. Execution evidence. The method’s instrumentation reported a token count consistent with its configured budget (Appendix A).

2. Prediction divergence from the unmodified backbone. Greedy decoding makes the pipeline a deterministic function of (weights, frames, prompt): frame indices come from a fixed linear spacing, the vision tower is frozen, token choice is arg max. Re-running an unchanged configuration reproduces all 8,052 predictions exactly — confirmed on three independent run pairs, each diverging on 0 of 8,052. Against that zero baseline, a method that discards or merges most of the visual input and produces zero differing predictions cannot have executed.

The second signal is not redundant. Four ports passed every other check and failed it: an early FastV integration whose pruning call was a stub; a PruneVID port to LLaVA-OV whose discard mask was dropped before reaching the model; and FlashVID and AIM on Qwen3-VL, which logged a correct budget (`keep=1750 (15.0%)`) while emitting byte-identical output, because `attention_mask` is `None` under this model’s default attention implementation and the mask-dependent path never ran. A fifth, STTM on Qwen3-VL, failed because that model interleaves timestamp text tokens between frames — the 11,664 visual tokens occupy an 11,784-wide span broken by 15 gaps, and a patch assuming one contiguous block did nothing. All five would have entered the results tables as working methods under an execution-log check alone.

Measured token budgets

Table 9: Measured visual-token budget on Qwen3-VL-8B, read from each method’s execution log rather than its CLI flag. Input is 11,664 visual tokens at 32 frames. Nominal 15% describes five of ten methods.

Method	Reduction stage	Measured budget
HoliTom	pre-LLM segment retain + in-LLM merge	7.5% (875 tokens)
STTM	quadtree spatio-temporal merge	9.0% (1,050)
FastV	in-LLM attention rerank, layer 2	15.0% (1,750)
FlashVID	pre-LLM temporal-segment merge	15.0% (1,750)
AIM	bipartite merge + PageRank prune	15.0% (1,750)
VisionZip (contextual)	key-similarity merge in vision tower	15.0% (1,750)
DyCoke	temporal merge + KV prune, layer 3	49.0% (5,716)
PruneVID	DPC-KNN cluster merge, layer 10	50.0% (5,832)
MDP3	frame selection (DPP)	8 of 32 frames
VideoITG	grounded frame selection	32 of 512 sampled

Provenance

Every figure in the results tables and all four figures were recomputed for this report directly from each run’s per-sample prediction records, not transcribed from prior documentation. Recomputed accuracy agreed with each run’s stored summary to within 0.01 points across all available run directories, and independently reproduced the documented divergence counts, paired win/loss counts

and per-category figures.

An earlier revision omitted four LLaVA-OV results — MDP3, AIM, VideoITG and STTM — because their run directories were absent from the local cache and only non-comparable sampled-decoding variants were present. Those four runs have since been retrieved and independently recomputed on the same basis as every other row; they now appear in Tables 1, 3 and 10. The DyTo token-matched control (§2.9) remains cited rather than recomputed, as no baseline run exists on its backbone.

Six Repetition Count cells land on an exact rounding tie: HoliTom and PruneVID-OV at $109/400 = 27.25\%$, DyCoke and FastV at $101/400 = 25.25\%$, AIM at $89/400 = 22.25\%$ and STTM at $115/400 = 28.75\%$. This report rounds half away from zero throughout, giving 27.3, 25.3, 22.3 and 28.8. Python’s default banker’s rounding disagrees on every one of them, so recomputation scripts must not be used to “correct” these cells without changing the stated convention first.

B Paired-prediction statistics

Differ counts predictions unequal to the backbone’s over all 8,052 rows (accuracy uses the 4,018 scoreable subset); 0 would indicate a silent no-op. *Fixed* and *broke* are the McNemar discordant pairs on scoreable items.

Divergence volume does not predict damage, and the two must not be conflated. STTM on LLaVA-OV changes 4,859 predictions — more than FastV’s 4,605 — yet loses 0.94 points and is not significant, because its changes fall almost evenly in both directions (329 fixed, 367 broken). FastV changes a comparable number destructively (475 fixed, 1,113 broken). Divergence establishes only that a method executed; the discordant pairs establish what it cost.

Table 10: Paired-prediction statistics against each backbone.

Method	Differ	Fixed	Broke	χ^2	Significant
<i>LLaVA-OV-7B</i>					
DyCoke	1,031	170	142	2.34	no
FlashVID	1,610	268	242	1.23	no
HoliTom	1,838	299	280	0.56	no
MDP3	1,452	246	230	0.47	no
VideoITG	1,193	192	184	0.13	no
AIM	1,406	227	219	0.11	no
STTM	4,859	329	367	1.97	no
PruneVID-OV	4,536	473	1,054	220.30	yes
FastV	4,605	475	1,113	255.52	yes
<i>Qwen3-VL-8B</i>					
PruneVID	1,074	56	70	1.34	no
DyCoke	1,048	83	110	3.50	no
HoliTom	1,657	131	219	21.63	yes
MDP3	1,626	159	274	30.01	yes
FastV	2,029	149	290	44.65	yes
VisionZip (contextual)	2,035	191	340	41.25	yes
STTM	2,095	170	389	85.02	yes
FlashVID	2,625	178	414	93.29	yes
VideoITG	2,522	206	454	92.44	yes
AIM	2,861	194	457	105.44	yes
<i>Backbone vs. backbone</i>					
Qwen3-VL-8B vs. LLaVA-OV-7B	7,081	942	546	127.47	yes

C Limitations

Scope. All results are on MotionBench and two backbones. Nothing here generalizes to other video benchmarks; the per-category analysis of §2.5 in fact predicts that these methods will look better on benchmarks weighted toward appearance and object recognition.

Single measurement per cell. Greedy decoding makes each configuration exactly reproducible, so there is one measurement per cell and no run-to-run variance estimate. Reported uncertainty is binomial, not empirical — which is why the ± 1.54 -point floor is applied strictly.

Standardization is not paper fidelity. FastV’s published default is $r = 0.5$; the figures here are FastV *at* 10–25%, well outside its validated range, and should not be read as characterizing the method as published. Since methods without a retention knob keep their defaults, the cross-method comparison is at a common nominal setting and not at equal compute (Appendix A).

Determinism is conditional. Bit-identical reproduction was verified across nodes of the same GPU generation, same software stack, batch size 1. Different GPU generations may perturb near-tied logits; that comparison was not run. Replicators on other hardware should expect close, not identical, numbers.

Zero divergence is strong but not airtight. A method pruning only tokens the model already ignored would execute correctly and produce zero divergence, and the gate would misclassify it. No such case was observed.

Scoring protocol. Accuracy depends on a letter-matching regular expression and on excluding the 4,034 NA items — roughly half the benchmark. Both are documented choices; neither has been

ablated.

D Open items

1. **Where the discard threshold falls.** §2.8 establishes that on LLaVA-OV a 15% budget costs about 15 points regardless of selection rule, while 100% retention through the same code path is intact. The transition between those two points was not measured. FastV’s own published default, $r = 0.5$, sits inside the unmeasured interval, so this report cannot say how the method behaves at its specified operating point on this backbone.
2. **Whether the threshold is backbone-specific.** No equivalent selection experiment was run on Qwen3-VL, where the same methods respond smoothly to budget (§2.6) and show no collapse. The contrast is suggestive but untested.
3. **Sampled-decoding answer distribution.** A companion study under temperature = 0.7, top- $p = 0.9$ (not comparable to any gated result, since sampling defeats the divergence check) is missing its FlashVID row; that run is in flight.